

Received 24 July 2026, accepted 21 August 2026, date of publication 27 August 2026, date of current version 11 September 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3727721

RESEARCH ARTICLE

Intention Recognition for Digital Forensics: With Agentic AI

YIDNEKACHEW WORKU KASSA¹, ELEFELIOUS GETACHEW BELAY², AND JOSHUA I. JAMES³

¹School of Information Technology and Engineering, College of Technology and Built Environment, Addis Ababa University, Addis Ababa, Ethiopia

²Center of Artificial Intelligence and Robotics, Institute of Advanced Science and Technology, Addis Ababa University, Addis Ababa, Ethiopia

³DFIR Science LLC, Champaign, IL 61821, USA

Corresponding author: Elefelious Getachew Belay (elefelious.getachew@aau.edu.et)

ABSTRACT Digital forensic investigations are increasingly challenged by the growing volume, velocity, and complexity of digital evidence, which can delay investigations and overwhelm forensic resources. While existing approaches have advanced the reconstruction of technical events, they provide limited support for systematically reasoning about criminal intent, a critical factor in legal attribution and prosecution. This study introduces the Intention Recognition for Digital Forensics (IRDF) framework, a legally grounded and implementation-independent model that integrates technical evidence interpretation with intent-based legal reasoning. The framework formally represents crimes, intent categories, evidentiary elements, actions, and states, providing a structured foundation for intention-aware forensic analysis. To evaluate its feasibility under controlled experimental conditions, the framework is realized through an agentic AI implementation supported by a structured Criminal Knowledge Base (CKB). The implementation was evaluated through a scenario-based experimental study spanning phishing, denial-of-service, and ransomware investigations, multiple intent categories, repeated executions, and three Large Language Models (OpenAI, Gemini, and DeepSeek), under both knowledge-guided and baseline conditions. The results indicate that structured knowledge guidance improved intent-classification accuracy across all evaluated models, increasing accuracy from 33.3% to 66.7%, and 44.4% to 75.0% and 47.2% to 69.4% for DeepSeek, Gemini, and OpenAI, respectively. At the same time, unsupported intent inflation was substantially reduced from 66.7% to 22.2%, and 55.6% to 11.1%, and 44.4% to 19.4%, respectively, resulting in more disciplined, explainable, and legally aligned reasoning behavior across models. The findings demonstrate the potential of knowledge-guided agentic AI to function as an intelligent forensic assistant capable of supporting increasingly complex investigations while maintaining necessary human oversight. The study contributes both a reusable formal framework for intention-aware digital forensic reasoning and an empirically evaluated agentic AI realization that demonstrates its potential to practical feasibility.

INDEX TERMS Cybercrime, digital forensics, intention recognition, formal model, agentic AI, knowledge-guided agents.

I. INTRODUCTION

The exponential growth of cyberspace, coupled with increasing technological dependence, has created an environment conducive to the proliferation of cybercrime [1], [2]. Cybercrime now encompasses not only direct attacks on digital infrastructure, but also traditional crimes facilitated through digital technologies, increasingly blurring the distinction between cyber and conventional offenses [3], [4], [5].

The associate editor coordinating the review of this manuscript and approving it for publication was Anandakumar Haldorai¹.

As a result, Digital Forensics (DF) laboratories face growing investigative pressure, driven not only by increasing case volumes, but also by the scale, heterogeneity, and complexity of digital evidence requiring analysis [6], [7]. Digital investigation is therefore increasingly recognized as a big-data and cognitive analysis challenge, where conventional human-centered workflows struggle to scale without intelligent automation [8], [9], [10].

Various strategies have been proposed to address scalability challenges in DF, including data reduction, triage, data mining, and intelligence-driven approaches [10]. While these

methods improve evidence handling efficiency, they primarily focus on technical processing rather than higher-order investigative reasoning. DFIs ultimately seek not only to establish what occurred, but also to support attribution, culpability assessment, and legally meaningful conclusions. This makes Intention Recognition (IR) particularly more relevant. Existing IR research spans multiple domains and employs diverse techniques, in DF probabilistic, logic-based, and learning-based approaches are mostly utilized [11], [12], [13], [24]. However, these approaches predominantly interpret intent from a technical or behavioral perspective, typically to predict attacker objectives or anticipate malicious actions.

A critical gap remains in applying IR to DF from a legally grounded perspective. In forensic investigations, technically suspicious behavior alone does not necessarily establish criminal culpability; legal interpretation requires structured reasoning about intent, including distinctions such as purposeful, knowing, reckless, or negligent (ignorance) conduct. Despite advances in both DF and cyber IR, this legal dimension remains largely underexplored in AI-assisted forensic investigation.

To address this gap, this study proposes IR based framework for digital forensics investigation (DFI) that integrates technical forensic reasoning with legally grounded intent analysis. Furthermore, the study implements the framework using an Agentic-AI approach and demonstrate it using different scenarios. Unlike conventional cyber IR approaches that focus on prediction or detection, the proposed framework repositions IR as an investigative reasoning mechanism capable of supporting DF interpretation and legal decision support. The realization of the framework combines modular agentic reasoning with a structured Criminal Knowledge Base (CKB) that links technical evidence to legal constructs, enabling traceable and jurisdiction-aware forensic reasoning while maintaining human oversight.

The proposed architecture is conceptually end-to-end, spanning identification, examination, analysis, and reporting; however, this study empirically focuses on the analysis and recommendation phases, where intention reasoning plays its most explicit operational role. The framework is evaluated through different simulated forensic scenarios spanning phishing, distributed denial-of-service (DDoS), and ransomware cases, with repeated comparative experiments across multiple Large Language Models (LLMs) to assess consistency, reasoning behavior, and the impact of knowledge grounding.

The remainder of this paper is organized as follows: Section II reviews the background and related works, covering DF, IR, agentic AI for DFI, and the research gap addressed by this study. Section III presents the research methodology. Section IV introduces the formal foundations of the IR modeling framework, including the formal definitions of crime, criminal intent, and intent types. Section V presents the agentic AI-based IR model, including the architectural rationale, integrated model, model components, CKB, and

operationalization of the formal model. Section VI presents the scenario-based evaluation and analysis, including the evaluation objectives, experimental design, and comparative findings from the analysis and recommendation agents. Section VII discusses the broader implications of the findings and the limitations of the study. Finally, Section VIII concludes the paper and outlines future research directions.

II. BACKGROUND AND RELATED WORKS

A. DIGITAL FORENSICS FOUNDATIONS

Digital Forensics is a specialized field of forensic science that focuses on the recovery and investigation of digital technologies, often in connection with computer crime [14], [15]. The practice involves the application of scientific principles and technological methods by considering legal requirements, to collect, analyze, and present data from digital sources [15], [16]. It plays a crucial role in solving crimes that involve digital technologies, ensuring that digital evidence is handled in a legally admissible way.

The essence of DF lies in its ability to establish the truth about digital events or activities. It applies a rigorous, scientific approach to the digital domain to uncover facts that can have real-world consequences, combining elements of law, computer science, and ethics [14], [16], [17]. DF provides crucial evidence in legal proceedings, enhances cybersecurity, aids in incident response, ensures accountability, helps prevent future attacks, recovers data, contributes to cybersecurity intelligence, and builds trust in digital systems.

DF faces several challenges, including the overwhelming volume of data, strong encryption, data destruction, device diversity, cloud forensics, legal and ethical issues, tool efficacy, training and resources, rapid technological change and the complexity of cases [6], [17], [18], [19], [20]. Existing automation has largely focused on technical evidence handling, including acquisition, parsing, filtering, and triage [8], [9]. However, the cognitively intensive reasoning tasks of evidence interpretation, intent assessment, and investigative decision support remain comparatively underexplored, motivating the need for more intelligent forensic assistance.

B. INTENTION RECOGNITION FOUNDATION

Intention Recognition refers to the process of inferring an actor's objectives, motivations, or likely future actions based on observable behavior, contextual evidence, or historical patterns [11]. IR has been applied across multiple domains, including cybersecurity, behavioral analytics, social media analysis, and crime prediction, using diverse approaches such as probabilistic reasoning, logic-based models, machine learning, and deep learning techniques [12], [13], [24]. In cybersecurity, IR is commonly used to anticipate attacker objectives, recognize malicious action sequences, and support proactive defensive responses.

However, the application of IR in DFs presents a materially different challenge. Unlike operational cybersecurity, which is largely predictive or preventive, DFI is retrospective,

evidentiary, and legally consequential. The objective is not merely to infer likely attacker behavior, but to interpret observed actions and state changes within an evidentiary framework that supports attribution and legally meaningful conclusions. In this context, technically inferred intent alone is insufficient, as similar observable actions may reflect malicious conduct, negligence, experimentation, misconfiguration, or legitimate administrative activity.

Since DFI is a highly sensitive task that requires meticulous attention to detail and a cautious approach, the application of IR in this domain necessitates a deep understanding of the context as well as precise specification of the requirements. To effectively define and clarify these requirements, the use of formal methods presents an ideal solution, offering rigorous, structured approach to ensure accuracy and reliability in requirement specification process [27], [28].

C. AGENTIC AI FOR DFI

Recent advances in LLMs have significantly expanded the potential of AI-assisted analytical reasoning, enabling capabilities such as contextual interpretation, structured reasoning, evidence synthesis, and complex decision support [65], [67], [68], [69]. Building on these capabilities, agentic AI extends beyond single-prompt interaction by decomposing complex tasks into specialized autonomous or semi-autonomous reasoning roles coordinated through modular orchestration mechanisms [70], [71], [72]. This architectural shift is particularly relevant for complex investigative domains where reasoning tasks are multi-stage, iterative, and context dependent.

DFI presents a compelling application domain for such intelligent assistance. Traditional forensic automation primarily focuses on technical artifact acquisition, parsing, filtering, indexing, and evidence management [9], [10], [15], [76]. However, digital investigations also require substantial higher-order reasoning, including event reconstruction, evidence correlation, hypothesis generation and testing, attribution assessment, and legally meaningful reporting [16], [77], [80]. Recent research has begun exploring AI-assisted support in DF contexts, including machine learning for evidence classification, anomaly detection, malware analysis, and forensic triage [6], [18], [20]. Emerging LLM-based studies further suggest potential for broader analytical assistance, although current applications largely remain task-specific or technically focused rather than supporting end-to-end investigative reasoning.

Despite this promise, generic AI reasoning may remain insufficient for justice-sensitive forensic environments. Unlike conventional analytical domains, DF outputs may directly influence investigative direction, prosecutorial escalation, evidentiary interpretation, and judicial outcomes. Consequently, forensic AI requires stronger evidentiary discipline, explainability, accountability, and legally defensible reasoning than general analytical automation [73], [75], [81]. This motivates the need for specialized agentic architectures

that combine scalable AI reasoning with explicit forensic and legal constraints. The present work builds on this direction by proposing a legally grounded agentic framework for DF intention based reasoning.

D. SUMMARY OF RELATED WORK AND RESEARCH GAP

Existing research relevant to this study can be broadly grouped into two streams. The first focuses on IR, which has been extensively investigated across diverse cybercrime-related domains, including Advanced Persistent Threats (APT), mobile security, AI security, network security, physical security, and social media, using a variety of domain-specific analytical approaches [21]. The second focuses on the application of AI, particularly LLMs and agentic AI, to support digital forensic investigation. While both streams have advanced significantly in recent years, limited attention has been given to integrating legally grounded IR within AI-assisted DFI, leaving an important research gap that this study seeks to address.

Several studies have addressed cyberattack IR using varied technical approaches. Ahmed et al. [29] propose a similarity-based fuzzy min-max neural network for distinguishing general and specific cyberattack intentions. Kim et al. [30] use attack trees in Android environments to detect intentions before and after attacks, including smishing and backdoor threats. Cheng et al. [31] develop an ontology-driven framework for modeling APT intentions in IoT settings, while Pang et al. [32] focus on adversarial attack intent detection against deep neural networks.

IR has also been extensively explored in network security. Mirsky et al. [33] introduce metric-based algorithms for goal recognition, while Chen et al. [34] employ attack graphs to infer complex multi-step attacker intentions. Li et al. [12] apply Hidden Markov Models (HMM) for multi-phase attack detection, whereas Zhang et al. [22] model attacker-defender interactions using game theory. Zhao et al. [35] apply heterogeneous attention networks for intruder intent prediction, and Kang et al. [36] propose an alert-driven classification system for attack IR.

Beyond cyberattack analysis, IR has also advanced in physical security and surveillance contexts. Hsu et al. [37] integrate access control, surveillance, and host defense for malicious activity detection, while Tang et al. [38] apply Graph Convolutional Networks to detect attack intentions in power systems. In video surveillance, Naval Gund et al. [13] develop a real-time weapon detection system for CCTV environments, Martinez-Mascorro et al. [39] use 3D Convolutional Neural Networks (CNNs) to identify shoplifting intent, and Bhugul et al. [25] propose a deep learning model for suspicious activity detection.

IR has also been explored in social media and communication platforms for criminal or malicious behavior detection. Mendonça et al. [78] develop a framework for detecting criminal intent in Portuguese social media slang, while Abarna et al. [24], Bokolo et al. [23], and Pandey et al. [40]

apply Natural Language Processing (NLP) and machine learning approaches to detect cyberharassment, malicious intent, and harmful behavioral patterns in social media content. Tsinganos et al. [26], [41] further extend this direction to persistent chat-based social engineering attack detection.

A common limitation across these studies is their predominant focus on technical intent recognition. While technically effective, they generally do not address intention within broader legal or evidentiary contexts, where intent interpretation must support attribution, culpability assessment, and legal decision-making. This observation is consistent with research in legal informatics and AI & Law, which has long emphasized the importance of structured legal knowledge, explainable reasoning, and evidence-based analytical support in legally consequential decisions [83], [84]. Our approach addresses this gap by integrating IR within a legally grounded DF framework.

Recent developments have also examined the integration of Agentic AI, particularly LLMs, into DF and legal workflows [42], [43], [44]. These studies highlight the growing potential of LLMs to support evidence analysis, entity recognition, knowledge reconstruction, legal search, and investigative assistance, while addressing persistent workload and procedural challenges in forensic practice.

More specifically, recent studies have begun applying LLMs to specialized DF tasks. Rahman et al. [45] combine open-source LLMs with vector databases for network traffic analysis and cyber incident reconstruction. Zhou et al. [46] propose an automated framework for extracting criminal insights from digital evidence networks through Forensic Intelligence Graphs (FIGs). Lira et al. [47] demonstrate the use of LLMs for large-scale network log interpretation and adaptive threat detection, while Kim et al. [48] explore LLM-assisted analysis of mobile messenger data for mobile forensics. Michelet et al. [49] further show the practical value of LLMs in assisting forensic report writing under expert supervision.

In all, these studies demonstrate the growing applicability of LLMs in DF workflows. However, existing approaches primarily apply LLMs to specific forensic tasks, whereas the proposed IRDF framework organizes specialized agents around a formally defined intention-recognition process and uses the a Knowledge Base to guide reasoning through structured legal and forensic knowledge. To the best of our knowledge, no prior work explicitly proposes an agentic LLM-based framework for legally grounded IR in DFIs. The present study addresses this gap by introducing such a framework to support analytical reasoning in DF contexts.

III. RESEARCH METHODOLOGY

This study adopts a Design Science Research (DSR) methodology, which is well suited for addressing practical problems through the design, development, and evaluation of innovative artifacts [66]. DSR is appropriate for this work

because the study aims to address a concrete challenge in DFI, namely the growing difficulty of interpreting large and complex digital evidence while maintaining legally meaningful reasoning about criminal intent. Following the DSR process, the study identifies the problem domain, defines the research objectives, develops a novel artifact, and evaluates its applicability through structured experimentation.

The primary research artifact is a legally grounded IR based DF framework, and an agentic AI based implementation of the framework. The artifact integrates DF reasoning with criminal intent interpretation by combining an agent-based analytical workflow with a structured CKB that captures legal intent categories, crime elements, technical indicators, and supporting evidentiary relationships. The framework operationalizes IR as an analytical reasoning process that links observable digital actions and states to legally interpretable intent classifications. While the broader conceptual model supports end-to-end DFI, the implementation focus of this study is on the analysis and recommendation phases, where intent reasoning is most directly applied.

Evaluation was conducted using a scenario-based experimental methodology to assess the applicability, consistency, and comparative behavior of the proposed framework under diverse investigative conditions. Multiple cybercrime scenarios were designed to represent distinct intent categories across different offense types. Experiments were executed across three LLMs with repeated runs under controlled prompting conditions, both with and without the CKB.

IV. IR MODELING FRAMEWORK: FORMAL FOUNDATIONS

This section establishes the formal foundations of the proposed IR framework by defining the core legal-technical constructs required for modeling criminal intent in DFI. The formalization is designed as a technology-agnostic conceptual model that provides a structured representation of crimes, intent categories, evidentiary elements, actions, and states in a machine-interpretable form. This abstraction is intended to support automated reasoning across different implementation paradigms, including existing and future computational approaches. The following subsections depicts formal definitions.

A. FORMAL DEFINITION OF CRIME

Definition 1: Crime is an action or omission that constitutes an offense and is punishable by law. [50], [51], [52], [53], [54]

This definition encompasses three essential elements: first, it identifies the specific action or omission attributed to the suspect; second, it recognizes that such actions or omissions are legally classified as offenses against individuals or society; and third, it stipulates that the offense is punishable under the law, commensurate with the degree of culpability. Thus, the concept of crime hinges on the simultaneous fulfillment of these three conditions.

Let

Sets:

- Let $\mathcal{C}$ = Set of all crimes.
- Set $\mathcal{A}$ = Set of all possible actions and inactions.
- Set $\mathcal{O}$ = Set of all offenses.
- Set $\mathcal{P}$ = Set of all punishments.

Predicates:

- $\text{Constitutes}(a, o)$: Specifies that Action (or Inaction) a constitutes an Offense o . This can be expressed as (1)

$$\exists o \in \mathcal{O} \mid \text{Action}(a) \wedge \text{Constitutes}(a, o) \quad (1)$$

- $\text{Punishable}(o, p)$: Indicates that offense o is punishable by law with punishment p . This can be expressed as (2)

$$\exists p \in \mathcal{P} \mid \text{Offense}(o) \wedge \text{Punishable}(o, p) \quad (2)$$

- Crime: The definition of crime can be expressed as (3)

$$\begin{aligned} \forall c \text{Crime}(c) \Leftrightarrow & \text{Action}(a) \wedge \text{Offense}(o) \\ & \wedge \text{Constitutes}(c, a) \wedge \text{Constitutes}(a, o) \\ & \wedge \text{Punishable}(o, p) \end{aligned} \quad (3)$$

A crime constitutes of entity a that is either an action or omission, which in turn constitutes an offense o , and the offense is punishable by law with punishment p .

B. CRIMINAL INTENT

Criminal intent, also termed mens rea, denotes the specific mental state or mindset that is legally requisite to substantiate a conviction for a criminal offense [55]. It encapsulates the defendant's state of mind at the time of committing the alleged criminal act, and it is crucial in determining culpability [55], [56], [57].

Criminal intent is a fundamental element that serves to differentiate between varying degrees of criminal responsibility based on the defendant's psychological disposition towards their conduct and its potential consequences [55], [58], [59].

To establish criminal intent, the prosecution must demonstrate that the defendant possessed a requisite level of mental awareness or purposefulness regarding their actions and the outcomes thereof. This mental state can vary across different crimes depending on the nature of the offense and the legal standards applicable [55], [58]. According to the Modal Penal Code, and Ethiopian computer crime proclamation, criminal intent can be categorized as Purposely, Knowingly, Recklessly and Ignorantly [54], [55]. the following are the formal definitions of Intent and Intent Types:

Definition 2: A suspect has **Criminal Intent** when the evidence shows that the suspect commits a crime and the criminal action (or inaction) matches the elements required for the specific level of intent defined by the law [55]. Formally, continuing from the above:

Let

- E : Set of all evidence,
- $I = \{I_p, I_k, I_r, I_i\}$: A finite set specifying the four levels of intention or mens rea, where:

- I_p represents Purposely: Acting with the intention to achieve a specific result.
- I_k represents Knowingly: Acting with awareness that a result is practically certain to follow.
- I_r represents Recklessly: Acting with disregard for a substantial and unjustifiable risk.
- I_i represents Ignorantly: Acting without knowledge of the nature of the act or its consequences.

- S : Set of all suspects involved in the criminal activities.
- $\text{Commits}(s, c, e)$: Specifies that the evidence e shows that the suspect s has committed the crime c . This can be expressed as (4)

$$\exists e \in E \mid \text{Suspect}(s) \wedge \text{Crime}(c) \wedge \text{Commits}(s, c, e) \quad (4)$$

- $\text{IsAdequate}(e, l, i)$: Specifies that the evidence e adequately satisfies the expected elements by the law l for the intent level i . This can be expressed as (5)

$$\begin{aligned} \exists e \in E \mid & \text{LawElements}(l) \\ & \wedge \text{IntentLevel}(i) \wedge \text{IsAdequate}(e, l, i) \end{aligned} \quad (5)$$

- $\text{CriminalIntent}(s, c, i)$: defines whether a suspect s has the criminal intent elements necessary for a crime c with intention i . For criminal intent to be established, the suspect must both commit the crime c and have the specific intention level i . Thus, criminal intent is true if and only if the evidence shows that the suspect has committed the crime and adequately satisfies the expected elements of the intention level i according to the law. This can be expressed as (6)

$$\begin{aligned} \forall s, c, i \text{CriminalIntent}(s, c, i) \Leftrightarrow & \text{Suspect}(s) \\ & \wedge \text{Crime}(c) \wedge \text{Intent}(i) \wedge \exists e, l \text{Evidence}(e) \\ & \wedge \text{LawElements}(l) \wedge \text{Commits}(s, c, e) \wedge \\ & \text{IsAdequate}(e, l, i) \end{aligned} \quad (6)$$

- Alternatively:

$$\begin{aligned} \forall s, c, i \text{CriminalIntent}(s, c, i, l) \Leftrightarrow & \text{Suspect}(s) \\ & \wedge \text{Crime}(c) \wedge \text{Intent}(i) \wedge \text{LawElement}(l) \\ & \wedge \exists e \text{Evidence}(e) \wedge \text{Commits}(s, c, e) \wedge \\ & \text{IsAdequate}(e, l, i) \end{aligned} \quad (7)$$

This definition (7) is equivalent to the above definition, however, it gives us the flexibility to reuse it in defining each type of intent based on the relation to the required intent elements by the law.

C. INTENT TYPES

The four types of intents can be formally defined as follows, continuing from above:

- Z : Set of all Outcomes.

1) Purposeful Intent:

Definition 3: A suspect exhibits purposeful intent if there exists a clear and deliberate goal to commit the

crime, and their actions are aimed at achieving this goal [55].

AimFor(s, z): Specifies that the suspect s aims for the outcome z . This can be expressed as (8)

$$\exists z \in Z \mid \text{Suspect}(s) \wedge \text{AimsFor}(s, z) \quad (8)$$

Purposely(s, c): Specifies that the suspect s has purposefully aim for the outcome of Crime c . This can be expressed as (9)

$$\forall s, c \text{ Purposely}(s, c) \Leftrightarrow \exists z \in Z \\ \mid \text{CriminalIntent}(s, c, I_p, \text{AimsFor}(s, z)) \quad (9)$$

There must be specific goals or plans (e.g., to cause harm) and the suspect's actions must be supported by evidence that shows these actions are aimed at achieving these goals.

2) Knowing Intent:

Definition 4: A suspect exhibits knowing intent if they are aware that their actions will likely contribute to the crime, even if they do not have the direct goal to commit the crime [55].

AwareOf(s, z): Specifies that the suspect s is aware of the outcome z . This can be expressed as (10)

$$\exists z \in Z \mid \text{Suspect}(s) \wedge \text{AwareOf}(s, z) \quad (10)$$

Knowingly(s, c): Specifies that suspect s is aware that their actions will almost certainly result in a specific outcome when committing crime c . This can be expressed as (11)

$$\forall s, c \text{ Knowingly}(s, c) \Leftrightarrow \exists z \in Z \\ \mid \text{CriminalIntent}(s, c, I_k, \text{AwareOf}(s, z)) \quad (11)$$

Evidence must show that the suspect was aware or should have been aware of the potential consequences of their actions, even if they did not intend to commit the crime.

3) Reckless Intent

Definition 5: A suspect exhibits reckless intent if they consciously disregard a substantial risk that their actions will result in a crime [55].

DisregardRisk(s, z): Specifies that the suspect s disregards a substantial risk of the outcome z . This can be expressed as (12)

$$\exists z \in Z \mid \text{Suspect}(s) \wedge \text{DisregardRisk}(s, z) \quad (12)$$

Recklessly(s, c): denotes that suspect s disregards a substantial and unjustifiable risk of causing a specific outcome when committing crime c . This can be expressed as (13)

$$\forall s, c \text{ Recklessly}(s, c) \Leftrightarrow \exists z \in Z \\ \mid \text{CriminalIntent}(s, c, I_r, \text{DisregardRisk}(s, z)) \quad (13)$$

There must be evidence that the suspect was aware of significant risks associated with their actions and acted despite these risks.

4) Ignorant Intent

Definition 6: A suspect exhibits ignorant intent if they are unaware of the harmful nature of their actions and do not know that their actions could contribute to a crime [55].

UnawareOf(s, z): Specifies that the suspect s is unaware of the outcome z . This can be expressed as (14)

$$\exists z \in Z \mid \text{Suspect}(s) \wedge \text{UnawareOf}(s, z) \quad (14)$$

Ignorantly(s, c): denotes that suspect s is unaware of the nature of their actions or the possible consequences when committing crime c . This can be expressed as (15)

$$\forall s, c \text{ Ignorantly}(s, c) \Leftrightarrow \exists z \in Z \\ \mid \text{CriminalIntent}(s, c, I_i, \text{UnawareOf}(s, z)) \quad (15)$$

Evidence must show that the suspect was genuinely unaware of the harmful nature or potential criminal outcome of their actions.

One of the basic properties of these intent types is they are mutually exclusive that a suspect cannot be convicted of a crime in more than one intent level [55]. This property can formally be captured as (16)

$$\forall i \neq j (\neg(I_i \wedge I_j)) \quad (16)$$

specifying for all pairs of intents sequenced by i and j (where $i \neq j$), the intent levels I_i and I_j cannot both be true simultaneously. This is because the intents are somehow hierarchical in the sense that to prove a suspect is convinced of a crime at some higher level of intent (like purposely intent), the prosecutor needs more additional evidence than lower level intent (like knowingly intent).

V. INTENTION RECOGNITION BASED DIGITAL FORENSICS (IRDF) FRAMEWORK

This section presents the proposed IRDF framework, first, introduces the formal model that defines the conceptual and analytical foundations of the approach, and subsequently describes its implementation using an agentic AI architecture.

A. IRDF CONCEPTUAL FRAMEWORK

The conceptual framework is designed to enhance investigations by utilizing the concepts of IR with component based modular architecture. As shown below, Figure 1, it accepts as input both the digital evidence collected from the crime scene and a crime narration that provides contextual understanding. There are two coordinating components, Governance & Oversight and Review & Validation that dynamically guide and evaluate the investigation process. The core analytical components, namely Evidence Examination and Intention Recognition, leverage both internal knowledge and external DF tools to process and interpret evidence in relation to suspected criminal intent. These components interact with a structured database composed of a custom-built Knowledge

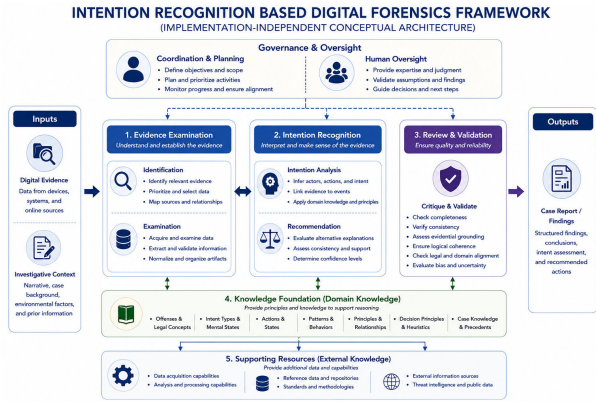


FIGURE 1. The IR high-level framework for DF.

Foundation and supplementary external forensic and threat intelligence databases. Based on this integrated reasoning pipeline, the model generates a DFI report, including intent-based interpretations and actionable recommendations to assist practitioners and legal experts in case resolution.

B. MODEL COMPONENTS

1) EVIDENCE EXAMINATION

The Evidence Examination component is central to filtering and examining the digital evidence to ensure only the relevant data is processed in subsequent stages of the investigation. The component is divided into two sub-components: Identification of Potential Evidence and Examination of Relevant Evidence.

a: IDENTIFICATION OF POTENTIAL EVIDENCE

The task of this sub-component is to filter and select potential evidence related to a specific crime, considering the details of the criminal act. This involves pulling the list of elements of the crime from the Knowledge Foundation and matching them against the overall digital evidence collected in relation to the suspect. The tasks of this sub component is highly related to the first tasks of the examination phase of the DFI process [15], [63].

- $\text{Related}(e, c, k)$: specifies that the evidence e is related to the crime c according to the crime knowledge repository k . This can be expressed as (17)

$$\begin{aligned} \exists e \in E \mid \text{Evidence}(e) \wedge \text{Crime}(c) \\ \wedge \text{KB}(k) \wedge \text{Related}(e, c, k) \end{aligned} \quad (17)$$

- $\text{PotentialEvidence}(p, c)$: denotes the set of potential evidence p , that is the subset of the overall evidence, which is filtered by applying the crime knowledge base related to the crime c . This can be expressed as (18)

$$\begin{aligned} \forall p, c \mid \text{PotentialEvidence}(p, c) \\ \Leftrightarrow \exists e, k \mid \text{Evidence}(e) \wedge p \in e \wedge \text{Crime}(c) \\ \wedge \text{KB}(k) \wedge \text{Related}(p, c, k) \end{aligned} \quad (18)$$

b: EXAMINATION: EXTRACTING RELEVANT EVIDENCE

This sub-component's task is to thoroughly examine the filtered potential evidence and assess its relevance for supporting the case. The goal is to examine the potential evidence to identify the relevant entries from logs, histories, files or any other potential evidence categories and to convert these raw evidence into a common format that can be processed by the Intention Recognition component. The evidence is analyzed for actions and states indicative of the suspect's behavior. The tasks of this sub-component are similar to the rest of the Examination phase of the DFI processes [15].

- $\text{RelatedEntry}(q, p, c, k)$: specifies that the entry q is recorded in the potential evidence p and it is related to crime c according to the knowledge base k . This can be expressed by (19)

$$\begin{aligned} \exists q \in Q, c, k \mid \text{PotentialEvidence}(p, c) \\ \wedge \text{RelatedEntry}(q, p, c, k) \end{aligned} \quad (19)$$

- $\text{Relevant}(r)$: specifies that the evidence r is a relevant entry in the potential evidence for the crime c according to the crime knowledge repository k . This can be expressed by (20)

$$\begin{aligned} \forall r \mid \text{Relevant}(r) \Leftrightarrow \exists r, p, c, k \mid \text{Crime}(c) \\ \wedge \text{KB}(k) \wedge \text{RelatedEntry}(r, p, c, k) \end{aligned} \quad (20)$$

2) INTENTION RECOGNITION

The IR component processes the evidence from the Evidence Examination component and determines the suspect's criminal intention. This component consists of two sub-components: Intention Analysis and Recommendation.

a: INTENTION ANALYSIS

The Intention Analysis sub-component is tasked with evaluating the suspect's involvement in various criminal activities based on the alignment of evidence with the knowledge base. This analysis involves assessing the extent of the suspect's engagement in crimes, as well as determining the potential for the suspect to face charges reflecting an appropriate degree of intent. The sub-module systematically calculates the likelihood of different levels of criminal intent, thereby providing a nuanced understanding of the suspect's criminal engagement. To do so the component utilizes the output of the Evidence Examination component to reconstruct the crime scene. The tasks of this component is highly related to the analysis phase of the DFI, however this component follows the IR perspective [15].

- $\text{Match}(r, k, c, i)$: specifies that each elements of the crime c at the intention level of i which are found in the crime Knowledge Base k has a match in the relevant evidence r . This can be expressed by (21)

$$\begin{aligned} \exists r, k, c, i \mid \text{Relevant}(r) \wedge \text{KB}(k) \wedge \text{Crime}(c) \\ \wedge \text{Intention}(i) \wedge \text{Match}(r, k, c, i) \end{aligned} \quad (21)$$

- $Culpable(s, c, i, r, k)$: specifies that the Suspect s is culpable of the Crime c at an intention level of i , given the match of the Relevant evidence r and the KB k . This can be expressed by (22)

$$\begin{aligned} \forall s, c, i \text{ Culpable}(s, c, i, r, k) &\Leftrightarrow \exists r, k \mid \text{Crime}(c) \\ &\wedge \text{Suspect}(s) \wedge \text{Intention}(i) \wedge \text{Relevant}(r) \\ &\wedge \text{KB}(k) \wedge \text{Match}(r, k, c, i) \end{aligned} \quad (22)$$

b: RECOMMENDATION

The Recommendation sub-component is responsible for guiding the investigation process by proposing optimal strategies based on the suspect's level of involvement. It evaluates the likelihood of uncovering additional evidence and suggests investigative pathways that align with the suspect's degree of engagement in the crime. This sub-module ensures that the investigative approach is both targeted and efficient, facilitating a thorough examination of the evidence and supporting the formulation of appropriate charges. The tasks of this sub component are mainly related to the reporting phase of the DFI, and this component emphasizes on the recommendations [15].

- $NeededEvidence(d)$: specifies that from the elements of the crime c at the intention level of i which are found in the crime Knowledge Base k , there are some elements d which are missing in the relevant evidence r . This can be expressed by (23)

$$\begin{aligned} \exists d \mid \text{Relevant}(r) \wedge \text{KB}(d) \wedge \text{Crime}(c) \\ \wedge \text{Intention}(i) \wedge \neg \text{Match}(r, d, c, i) \end{aligned} \quad (23)$$

- $RelatedToOtherCrime(r)$: specifies that the Relevant evidence r is also related to other Crime c' according to the KB k . This can be expressed by (24)

$$\begin{aligned} \forall r, c' \text{ RelatedToOtherCrime}(r, c') \\ \Leftrightarrow \exists r \mid \text{Relevant}(r) \wedge \text{KB}(k) \\ \wedge \text{Crime}(c') \wedge \text{Related}(r, c', k) \end{aligned} \quad (24)$$

To ensure practical applicability and seamless integration into existing DF workflows, the proposed framework is designed to interface with widely used forensic tools such as FTK Imager, The Sleuth Kit (TSK), Autopsy, and other industry-standard platforms. By interacting directly with these tools, the model is capable of accessing, parsing, and interpreting raw forensic data ranging from disk images and file system artifacts to log files and registry entries. Furthermore, the ability to communicate with these tools enhances the model's utility as an investigative assistant. The integration with the forensic toolchains facilitates a more streamlined and intelligent investigation process, ultimately improving both the speed and reliability of forensic analysis.

3) KNOWLEDGE FOUNDATION

The knowledge repository supports both the Evidence Examination and IR components processes. It contains a database of legal frameworks, crime definitions, investigator

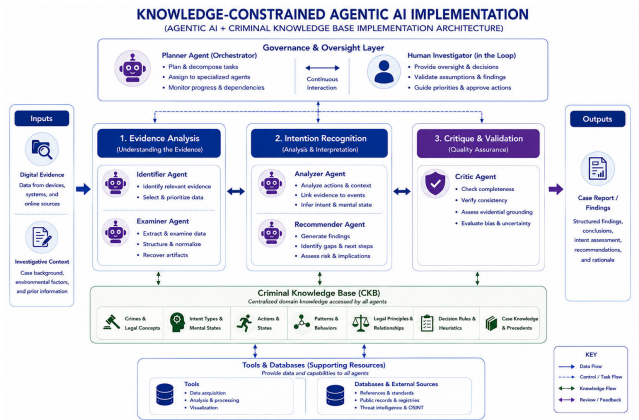


FIGURE 2. Agentic AI and CKB based implementation of the IRDF framework.

experiences, and DFs best practices, enabling effective analysis and interpretation of evidence. It provides the necessary information to identify and categorize evidence, as well as assist in determining the appropriate criminal intention level. Furthermore, to enrich the model's learning and inference capabilities, we better complement this core dataset with selected external resources as necessary, by employing different kinds of related databases such as the Solve-IT (Systematic Objective-based Listing of Various Established (Digital) Investigation Techniques) [62], the CASE (Cyber-investigation Analysis Standard Expression) ontology [79], and the MITRE ATT&CK framework [64].

C. AGENTIC AI IMPLEMENTATION OF THE IRDF FRAMEWORK

While the IRDF framework is defined as a formal, implementation-independent model, its practical application requires a computational realization capable of analyzing digital evidence, reasoning about criminal intent, and generating investigative recommendations. In this work, the framework is implemented using an agentic AI architecture in which the different sub-components are realized as specialized agents. The implementation is supported by a structured Criminal Knowledge-Base, the CKB, represented in JSON format, which operationalizes the formal concepts of crimes, intent categories, evidentiary elements, actions, and states.

Figure 2 illustrates the implementation architecture of the proposed framework. The formal components of the IRDF framework are realized as specialized agents, namely the Planner, Critic, Identifier, Examiner, Analysis, and Recommendation agents. Each agent is guided by task-specific prompts that define its responsibilities, objectives, expected outputs, and interaction patterns, enabling focused reasoning while maintaining consistency across the investigative process. The agents interact through a shared CKB, implemented in JSON format, which operationalizes the formal model by representing domain knowledge. The current implementation incorporates legal and investigative knowledge derived from Ethiopian and United States legal frameworks while

CRIMINAL KNOWLEDGE BASE (CKB) ARCHITECTURE
Simple, Structured Knowledge for Intention-Aware Digital Forensics

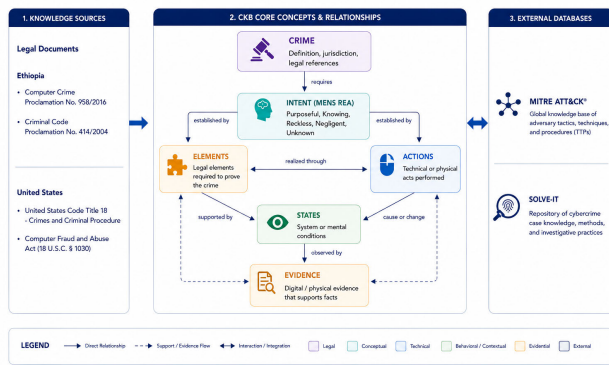


FIGURE 3. Architecture of the CKB.

remaining extensible to additional jurisdictions and crime categories.

The implementation follows a sequential and knowledge-guided workflow. The Planner coordinates the investigation by assigning tasks to the relevant agents, while the Identifier and Examiner agents collect and process digital evidence. The resulting evidence is analyzed by the Analysis agent, which consults the CKB to interpret observed actions and states, evaluate crime-specific evidentiary elements, and assess alternative intent classifications. This process operationalizes the formal IRDF model by linking digital artifacts to legally relevant crime elements and intent categories. The resulting analytical findings are then passed to the Recommendation agent to generate investigation reports and recommendations, while the Critic agent continuously evaluates intermediate and final outputs to support consistency, completeness, and evidentiary adequacy.

As shown in the CKB Architecture diagram, Figure 3, to enrich reasoning and improve interoperability, the CKB is complemented by selected external knowledge sources. Solve-IT [62] and the CASE ontology [79] provide structured representations that facilitate consistent interpretation and reasoning across heterogeneous evidence sources, while the MITRE ATT&CK framework [64] contributes domain knowledge on adversarial tactics and techniques. The relevant resources from these external databases are linked to the crime, or elements, or actions and states sections of the CKB. By combining structured legal knowledge, forensic evidence representations, and established cyber threat intelligence resources, the proposed implementation provides a knowledge-guided environment for IR that is both operationally relevant and adaptable to evolving DF challenges.

1) RATIONALE FOR AGENTIC AI WITH CKB ARCHITECTURE

The proposed implementation adopts a goal-oriented modular agentic architecture because IR in DFI requires the coordination of multiple analytical activities, including evidence examination, interpretation, intent assessment, and recommendation generation. A modular agent-based approach allows these activities to be separated into

TABLE 1. Operational mapping of formal model constructs.

Formal Construct	Operational Realization
Components	Agents
Crime	Structured crime definitions in CKB
Intent	Intent classification hierarchy in CKB
Element	Mandatory evidentiary requirements in CKB
Action	actions associated with elements in CKB
State	actions associated with elements in CKB
IR	The guidance intent rules in CKB

specialized reasoning functions while maintaining a coherent investigative objective. This aligns naturally with the structured and iterative nature of digital forensic investigations and provides a flexible foundation that can evolve alongside emerging cybercrime patterns, forensic techniques, and AI capabilities [60], [61].

The CKB was introduced to provide explicit legal and forensic knowledge that cannot be reliably inferred from language models alone. As depicted in Figure 3, by representing crimes, intents, evidentiary elements, actions, and states in a structured format, the CKB grounds reasoning in domain-specific concepts and supports consistent interpretation of evidence across cases and jurisdictions. The JSON representation was selected primarily for its simplicity, portability, extensibility, and compatibility with modern AI systems, facilitating future expansion of both legal knowledge and crime coverage.

The CKB was constructed by translating selected legal provisions into the formal structure defined by the proposed IR model. For each represented offense, the applicable intent requirements and evidentiary elements were identified from the corresponding legal definitions and mapped to technical actions, observable states, and supporting digital evidence relevant to DFI. The resulting entries were reviewed for structural consistency, ensuring that the encoded relationships among crimes, intents, elements, actions, and states remained traceable to their underlying legal definitions and consistent with the formal model. At this stage, this process provides structural and source-level validation of the CKB rather than formal legal validation.

The agentic architecture and the CKB provide complementary capabilities: the agents perform reasoning and task execution, while the CKB supplies the structured knowledge required to guide and constrain that reasoning. This combination supports explainable, adaptable, and knowledge-guided IR while maintaining the role of the investigator as the ultimate decision maker.

VI. SCENARIO-BASED EVALUATION AND ANALYSIS

A. EVALUATION OBJECTIVES

The evaluation aims to assess the effectiveness of the proposed knowledge-guided agentic DFI framework in supporting IR across cybercrime investigations. Since DF intention analysis involves both technical interpretation and legal reasoning, the evaluation involved not only intent classification accuracy but also on the reliability, consistency, and practical investigative utility of the framework.

The first objective is to assess whether integrating the CKB improves intent attribution compared to unguided LLM reasoning. Cybercrime investigations often involve technically suspicious artifacts that may arise from multiple plausible explanations, including malicious and non-malicious conduct. As a result, technical evidence alone may not reliably establish criminal intent. The evaluation therefore examines whether structured legal and technical knowledge grounding improves evidentiary interpretation by reducing inference inflation and encouraging evidence-sensitive intent attribution.

The second objective is to evaluate the comparative behavior and consistency of different LLMs under repeated execution. Since LLM outputs may vary across repeated runs due to probabilistic generation behavior [69], [74], the experiments assess the stability of intent classification and recommendation generation across multiple iterations and models.

The third objective is to evaluate the practical investigative value of the framework beyond classification alone. DFIs require not only analytical interpretation of evidence, but also defensible recommendations regarding evidentiary gaps, attribution strengthening, further investigative actions, and legal sufficiency. Accordingly, the evaluation includes both the analysis and recommendation agents to determine whether knowledge grounding improves the quality, proportionality, and defensibility of investigative outputs.

B. EXPERIMENTAL DESIGN

To evaluate the proposed framework under diverse cybercrime contexts, a structured experimental design was developed using controlled scenario-based simulations. Three representative cybercrime families were selected: phishing, DDoS, and ransomware. These crimes were chosen because they represent distinct technical and legal characteristics, including deception-oriented credential compromise, service availability disruption, and unauthorized data restriction, respectively.

For each crime family, four scenarios were designed to represent progressively different levels of criminal intent: purposeful, knowing, reckless, and ignorant conduct. To isolate the framework's IR capability, scenarios within each crime family maintained the same suspect and overall investigative context, while only the evidentiary conditions were varied. This controlled design allows comparative assessment of how changes in available evidence affect intent attribution while minimizing unrelated experimental variation.

The experimental workflow is illustrated in Figure 4. The evaluation began with preparation of the CKB, including legal definitions, technical indicators, intent decision rules, and mandatory/supporting evidentiary mappings. Standardized examiner-style forensic evidence packages were then prepared for each scenario as structured JSON inputs representing the outputs of DF examination. These evidence

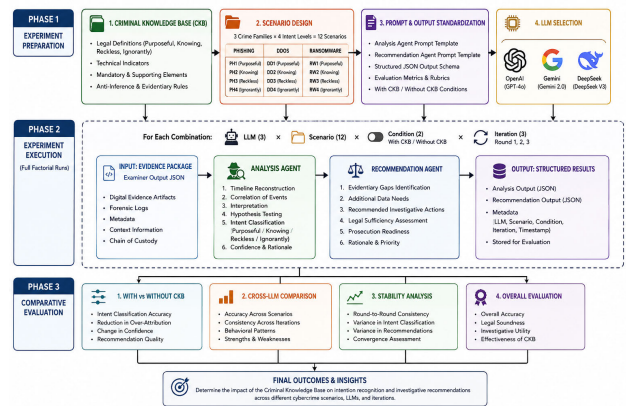


FIGURE 4. Experimental workflow for evaluating the proposed agentic DF intention-recognition framework across multiple cybercrime scenarios, LLMs, iterations, and knowledge-grounding conditions.

packages were processed by the analysis agent, which performed timeline reconstruction, evidence correlation, interpretation, hypothesis testing, and intent classification. The resulting outputs were subsequently processed by the recommendation agent to identify evidentiary gaps, propose additional investigative actions, assess attribution sufficiency, and evaluate legal readiness.

The experimental setup involved three LLMs: OpenAI (gpt-5.5), Gemini (gemini-3.1-pro-preview), and DeepSeek (deepseek-chat). Each scenario was executed under two conditions: (i) baseline execution without CKB grounding, and (ii) knowledge-guided execution with access to the CKB. To account for probabilistic variability in LLM behavior, each configuration was repeated three times.

The complete experimental design comprised 216 runs and 432 agent calls, derived from 3 crime families, 4 intent scenarios per family, 3 LLMs, 2 execution conditions (with and without CKB), and 3 repeated iterations per configuration. To support transparency and reproducibility, the implementation artifacts, experimental scenarios, prompts, CKB, the configurations, and supporting resources developed for this study have been made publicly available.¹

Quantitative evaluation used four metrics: classification accuracy, repeated-run stability, intent inflation rate, and under-attribution rate. Intent inflation measures cases where predicted culpability exceeded the expected scenario intent, while under-attribution captures overly conservative classifications.

C. RESULTS AND COMPARATIVE ANALYSIS

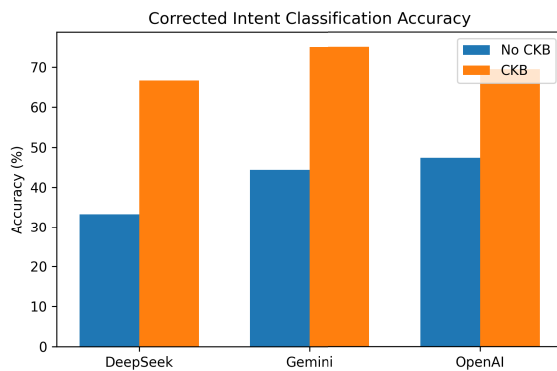
1) ANALYSIS AGENT FINDINGS

The analysis agent was evaluated from two complementary perspectives: (i) outcome-level performance based on final intent classification accuracy and consistency, and (ii) process-level reasoning quality across the forensic analysis pipeline, including timeline reconstruction, correlation, interpretation, hypothesis testing, and conclusion formation.

¹<https://github.com/YidnekWorku/IR-for-DF-with-Agents>

TABLE 2. Quantitative analysis-agent evaluation across baseline and knowledge-guided conditions.

Model	Condition	Accuracy	Stability	Intent Inflation	Under-Attribution
DeepSeek	Without CKB	33.3%	94.4%	66.7%	0.0%
DeepSeek	With CKB	66.7%	86.1%	22.2%	11.1%
Gemini	Without CKB	44.4%	94.4%	55.6%	0.0%
Gemini	With CKB	75.0%	91.7%	11.1%	13.9%
OpenAI	Without CKB	47.2%	83.3%	44.4%	8.3%
OpenAI	With CKB	69.4%	94.4%	19.4%	11.1%

**FIGURE 5.** Accuracy of intent classification.

a: OUTCOME-LEVEL QUANTITATIVE EVALUATION

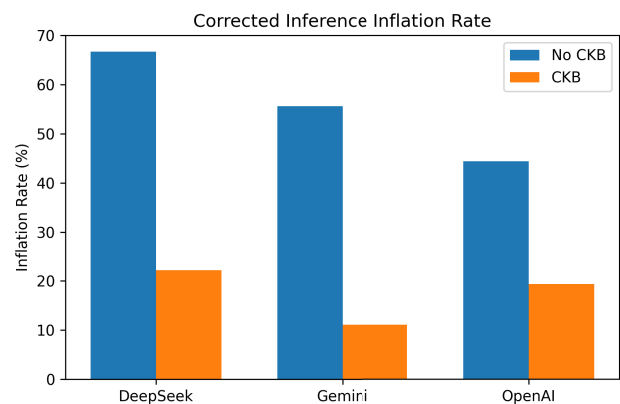
Table 2 presents the quantitative results across all evaluated LLMs under baseline and knowledge-guided conditions. Four metrics were used: classification accuracy, repeated-run stability, intent inflation rate, and under-attribution rate.

The quantitative results reveal several important findings. First, all models showed substantial improvement in intent classification accuracy under the knowledge-guided condition. DeepSeek improved from 33.3% to 66.7%, Gemini from 44.4% to 75.0%, and OpenAI from 47.2% to 69.4%. Gemini achieved the strongest overall accuracy under CKB-guided execution, while OpenAI demonstrated the strongest baseline performance.

Second, intent inflation emerged as one of the most important findings. Under baseline conditions, all models showed substantial escalation toward higher culpability than the scenario evidence supported. DeepSeek exhibited the strongest inflation tendency (66.7%), followed by Gemini (55.6%) and OpenAI (44.4%). This indicates that unguided LLM reasoning frequently interprets technically suspicious cyber artifacts as evidence of stronger criminal intent than warranted.

The introduction of the CKB substantially reduced this effect. Gemini showed the strongest reduction in inflation (55.6% to 11.1%), followed by DeepSeek (66.7% to 22.2%) and OpenAI (44.4% to 19.4%). This directly demonstrates the role of legal-technical knowledge grounding in constraining unsupported mens rea escalation. Figure 6 illustrates the reduction in intent inflation across models.

A particularly interesting observation concerns stability. Baseline repeated-run stability was already high

**FIGURE 6.** Reduction in intent inflation between baseline and CKB-guided conditions across evaluated LLMs.

for DeepSeek and Gemini, suggesting that the observed over-escalation was not merely stochastic generation noise but rather a systematic reasoning tendency. Knowledge grounding generally preserved or improved stability while significantly improving legal calibration.

b: PROCESS-LEVEL REASONING EVALUATION

While final classification outcomes are important, DF reasoning quality depends on the integrity of the analytical process itself. Accordingly, the outputs were further examined across the major reasoning stages of the analysis agent which are adopted from NIST.

Timeline reconstruction was consistently the strongest reasoning phase across all evaluated models. The models generally preserved event chronology, timestamps, source relationships, and actor associations correctly. This indicates that basic technical evidence extraction was not the primary weakness of the analysis agent.

The major weaknesses emerged in later reasoning stages. In correlation analysis, baseline models frequently linked technical artifacts too directly to criminal elements. For example, credential capture was often immediately correlated with purposeful fraud, traffic generation with deliberate service disruption, and encryption activity with extortion-oriented ransomware, even where alternative explanations remained plausible.

Interpretation emerged as the most problematic stage under baseline conditions. A recurring failure pattern was the conflation of technical intentionality with legal intentionality. In several scenarios, deliberate technical execution

TABLE 3. Process-level analysis-agent reasoning evaluation.

Reasoning Phase	Baseline Observation	CKB-Guided Observation
Timeline Reconstruction	Generally strong event extraction and chronological reconstruction across all models	Remained strong; minimal change since this phase is primarily technical
Correlation Analysis	Frequently aggressive evidence-to-crime mapping; technical artifacts often directly linked to criminal elements	More disciplined correlation with explicit legal/technical evidentiary mappings
Interpretation	Major over-escalation; technical deliberateness often interpreted as criminal intentionality	Improved proportionality through crime-specific evidentiary thresholds
Hypothesis Testing	Often shallow or performative; alternative hypotheses insufficiently explored	More structured comparison of competing intent hypotheses
Conclusion Formation	Frequently overconfident, insufficiently caveated, and prosecution-oriented	More evidence-proportionate and legally defensible conclusions

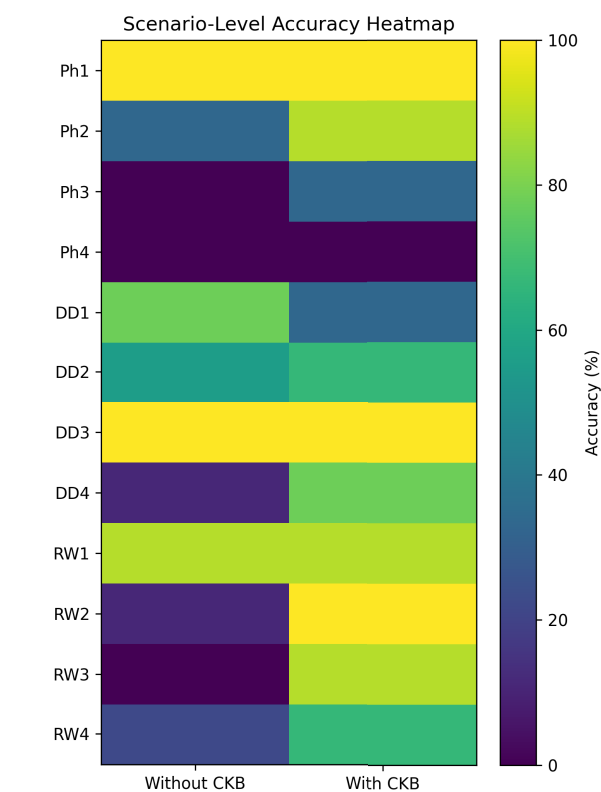


FIGURE 7. Process Level Reasoning Evaluation.

(e.g., launching a script, deploying an encryption tool, generating network traffic) was treated as sufficient evidence of purposeful criminal intent, despite lacking evidence for the prohibited criminal objective itself.

Hypothesis testing was similarly weak under baseline conditions. In many cases, alternative intent levels were nominally mentioned but not meaningfully evaluated. Instead, once a higher-intent hypothesis was adopted, competing hypotheses were often rejected without substantive evidentiary analysis. This pattern was particularly visible in ambiguous scenarios involving reckless experimentation or negligent administrative conduct.

TABLE 4. Comparative forensic reasoning behavior across LLMs.

Dimension	DeepSeek	Gemini	OpenAI
Baseline escalation tendency	Very High	High	Moderate
Baseline stability	High	High	Moderate
Interpretive proportionality	Weak	Moderate	Strong
Hypothesis-testing discipline	Weak	Moderate	Strong
CKB correction effect	Strong	Very Strong	Strong
Overall forensic reasoning quality	Moderate	Good	Best

The CKB improved these later reasoning stages substantially. By introducing explicit evidentiary thresholds, mandatory intent conditions, alternative explanation awareness, and anti-inference inflation constraints, the analysis agent became more disciplined in evaluating competing interpretations.

c: CROSS-MODEL REASONING BEHAVIOR

The three evaluated LLMs exhibited distinct forensic reasoning characteristics.

DeepSeek demonstrated the strongest baseline escalation tendency, frequently mapping technical suspiciousness directly to malicious intent. Its reasoning was often aggressive and less proportionate, although the CKB produced significant improvement.

Gemini exhibited the most stable baseline behavior, but this stability reflected consistently prosecution-oriented reasoning rather than correctness. Gemini frequently escalated toward purposeful classifications in ambiguous scenarios, making its strong correction under CKB guidance particularly significant.

OpenAI demonstrated the strongest baseline forensic reasoning quality. Compared to the other models, it more frequently acknowledged uncertainty, differentiated between intent levels, and avoided direct escalation to purposeful intent. OpenAI also showed the strongest hypothesis-testing discipline under knowledge-guided conditions.

d: CRIME-FAMILY OBSERVATIONS

Interesting differences also emerged across crime families.

Phishing scenarios produced the strongest baseline escalation across models. This appears to reflect semantic bias, where phishing artifacts strongly evoke fraud-related assumptions even in ambiguous or lower-culpability scenarios.

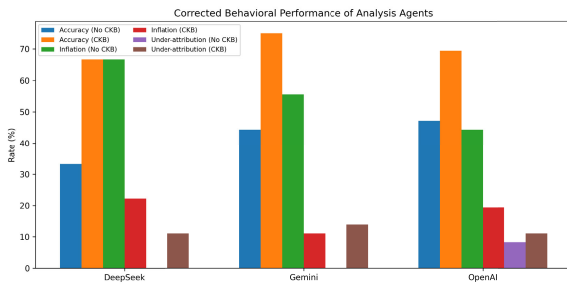


FIGURE 8. Cross-Model Reasoning Behavior.

DDoS scenarios proved the most difficult for intent discrimination. This is expected because similar technical actions may arise from malicious disruption, performance testing, experimentation, research, or administrative misuse. In some cases, the CKB introduced conservative corrections, suggesting that DDoS intent boundaries are particularly evidentially sensitive.

Ransomware scenarios demonstrated some of the strongest improvements under CKB guidance. The CKB objective-assurance approach proved especially effective in separating deliberate technical execution from extortion-oriented criminal purpose.

In summary, as discussed in the four different perspectives namely outcome level, process level, LLM behavior, and crime family observations, the analysis-agent evaluation demonstrates that the primary contribution of the CKB is not merely improved classification accuracy, but improved forensic reasoning discipline. The most substantial benefits emerged in interpretation, hypothesis testing, and conclusion formation, where legal-technical knowledge grounding constrained unsupported escalation and improved evidentiary proportionality.

2) RECOMMENDATION AGENT FINDINGS

While the analysis agent evaluates evidentiary interpretation and intent attribution, the recommendation agent addresses the operational decision-support dimension of DFI. Its role is to identify evidentiary gaps, propose additional investigative actions, assess attribution sufficiency, and evaluate legal readiness. Accordingly, the recommendation agent was evaluated from the perspective of practical forensic usefulness.

The recommendation outputs were examined across six dimensions: evidentiary gap awareness, attribution discipline, proportional escalation, alternative explanation sensitivity, investigative usefulness, and legal sufficiency judgment.

a: BASELINE RECOMMENDATION BEHAVIOR

A strong cross-model pattern emerged under baseline conditions. Recommendation outputs frequently reflected premature prosecutorial escalation, often assuming malicious culpability despite unresolved attribution uncertainty and incomplete evidentiary sufficiency.

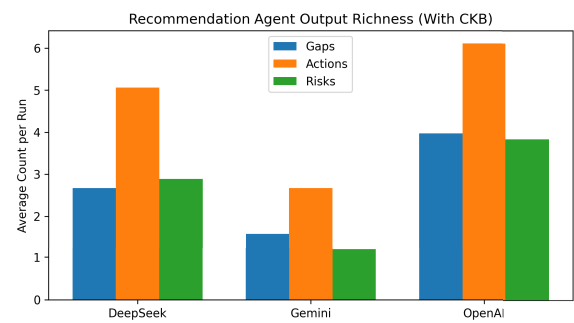


FIGURE 9. Baseline Recommendation Behavior.

In many cases, technically suspicious cyber artifacts triggered investigative recommendations that moved rapidly toward attribution pursuit, infrastructure seizure, escalation to legal enforcement actions, or prosecution-oriented evidence gathering, even where the underlying analysis remained evidentially ambiguous.

This baseline behavior is operationally significant because DFIs require proportional evidence-sensitive escalation. Premature prosecutorial assumptions can introduce confirmation bias, tunnel vision, weak attribution conclusions, and legally fragile investigative pathways.

b: MODEL-SPECIFIC RECOMMENDATION BEHAVIOR

The three evaluated LLMs exhibited distinct recommendation profiles.

DeepSeek demonstrated the most aggressive baseline recommendation behavior. Its outputs frequently assumed malicious culpability early, emphasized attribution pursuit, and recommended assertive escalation actions despite unresolved evidentiary uncertainty. While often operationally actionable, this behavior reflected weak proportional caution.

Gemini also showed strong baseline prosecution-oriented behavior, although with greater consistency and internal coherence than DeepSeek. Gemini frequently adopted a confident escalation posture, recommending legal progression even where ambiguity remained. This suggests that Gemini's baseline issue was less stochastic inconsistency and more systematic prosecutorial overconfidence.

OpenAI demonstrated the strongest baseline recommendation quality. Compared to the other models, it more frequently distinguished infrastructure attribution from actor attribution, acknowledged evidentiary insufficiency, and recommended proportionate investigative sequencing. Even without knowledge grounding, OpenAI more closely resembled real investigative reasoning than the other evaluated models.

c: IMPACT OF KNOWLEDGE GROUNDING

Knowledge grounding produced one of the strongest behavioral corrections observed in the study. Unlike the analysis agent, where improvements primarily affected interpretation and hypothesis testing, the recommendation agent exhibited a broader shift in investigative posture.

TABLE 5. Recommendation-agent evaluation dimensions.

Dimension	Evaluation Focus
Evidentiary Gap Awareness	Ability to identify missing evidence, unresolved uncertainties, and investigative insufficiencies
Attribution Discipline	Ability to distinguish infrastructure attribution from actor attribution and avoid premature suspect certainty
Proportional Escalation	Appropriateness of escalation toward legal or prosecutorial actions relative to evidentiary strength
Alternative Explanation Sensitivity	Recognition of ambiguity, competing hypotheses, or non-criminal explanations
Investigative Usefulness	Practical value and actionability of proposed next investigative steps
Legal Sufficiency Judgment	Quality of prosecution-readiness and evidentiary sufficiency assessment

TABLE 6. Comparative recommendation-agent behavior.

Dimension	DeepSeek	Gemini	OpenAI
Baseline caution	Low	Low	Moderate
Baseline prosecution bias	High	Very High	Moderate
Attribution discipline	Weak	Moderate	Strong
Gap identification	Moderate	Moderate	Strong
Alternative explanation sensitivity	Weak	Moderate	Strong
Investigative usefulness	Good	Moderate	Strong
Legal sufficiency discipline	Weak	Moderate	Strong
CKB behavioral correction	Strong	Very Strong	Strong
Overall recommendation quality	Moderate	Good	Best

Under CKB-guided execution, recommendations became substantially more evidence-sensitive. The models more consistently identified attribution weaknesses, evidentiary insufficiencies, unresolved intent ambiguity, and the need for further corroboration before legal escalation.

A particularly important improvement was the clearer distinction between infrastructure-level attribution and actor-level attribution. In baseline outputs, infrastructure evidence was often treated as sufficient proxy evidence for suspect culpability. Under CKB guidance, recommendation agents more frequently recognized that identifying a server, endpoint, or technical artifact does not establish the responsible actor.

Similarly, legal sufficiency assessments became more disciplined. Rather than assuming prosecution readiness from technical suspiciousness alone, CKB-guided recommendations more frequently emphasized the need for objective assurance, corroborative attribution evidence, stronger suspect linkage, and elimination of competing explanations.

d: INVESTIGATIVE POSTURE TRANSFORMATION

One of the strongest findings of the recommendation-agent evaluation is that knowledge grounding changed the overall investigative posture of the agent.

Baseline behavior can be characterized as prosecution-oriented:

- early escalation toward legal action,
- premature attribution assumptions,
- limited ambiguity tolerance,
- weak evidentiary caution.

CKB-guided behavior was more evidence-gap-oriented:

- identification of unresolved uncertainties,
- proportional investigative sequencing,
- stronger attribution discipline,
- clearer legal sufficiency thresholds.

This distinction is highly significant for DF practice. Real investigations require evidentiary discipline, not merely aggressive actionability. Recommendation systems that prematurely escalate weak cases may amplify investigative bias rather than improve forensic quality.

e: CRIME-FAMILY OBSERVATIONS

The recommendation-agent behavior also varied across crime families.

Phishing scenarios frequently triggered early fraud-oriented escalation, reflecting the strong semantic association between phishing artifacts and credential theft. This mirrors the analysis-agent inflation behavior.

DDoS scenarios produced more cautious recommendation behavior, particularly under knowledge-guided execution, likely because service disruption scenarios naturally admit broader ambiguity (e.g., testing, experimentation, misconfiguration, or malicious disruption).

Ransomware scenarios showed the strongest improvement in recommendation quality under CKB guidance, particularly where objective-assurance requirements forced clearer separation between deliberate technical interference and extortion-oriented criminal purpose.

To summarize, the recommendation-agent findings demonstrate that the contribution of knowledge grounding extends beyond analytical correctness into operational forensic decision support.

The key contribution is not simply that recommendations became “better,” but that the underlying investigative posture changed from premature prosecution-oriented escalation toward evidence-sensitive forensic decision support.

This is particularly important in DFIs, where operational recommendations can materially influence investigative direction, resource allocation, suspect treatment, and legal outcomes. Within the evaluated scenarios, the recommendation-agent results suggest that knowledge grounding improves the proportionality and evidentiary discipline of AI-assisted DF decision support, while indicating its potential to support more defensible and operationally useful investigative recommendations.

VII. DISCUSSION

DFI is increasingly confronted by a scalability challenge driven by the rapid growth of cybercrime, expanding technological complexity, and the resulting explosion in digital evidence volume [10], [77]. Traditional DF workflows

remain fundamentally human-centered, with automation largely focused on technical tooling for acquisition, parsing, and artifact analysis. While such tools remain essential, they do not adequately address the growing cognitive burden of interpreting evidence, correlating events, evaluating intent, and producing timely investigative conclusions [76], [80]. This study approached that broader challenge from the perspective of intent, which represents one of the central objectives of DFI. Beyond reconstructing technical events, DFIs ultimately support attribution, culpability assessment, and legally meaningful conclusions. Existing IR research has contributed significantly to technical cybersecurity applications, but its use in DFs has largely remained technically focused. By introducing legally grounded intention reasoning into forensic analysis, this work explores a pathway toward more intelligent and scalable investigative assistance.

The simulation findings suggest that LLMs can meaningfully contribute to this direction, but not as generic reasoning engines alone. The strong performance observed in timeline reconstruction across models indicates that contemporary LLMs are already capable of handling technically complex evidence synthesis tasks, consistent with emerging evidence of their reasoning utility in analytical domains [68], [69]. However, the analysis also revealed systematic weaknesses in higher-order forensic reasoning, particularly in interpretation, hypothesis testing, and conclusion formation, where baseline models frequently escalated technically suspicious behavior into unsupported higher culpability conclusions. This suggests that AI support for DFs should not be viewed simply as a more capable automation tool, but as a reasoning assistant requiring architectural specialization. Agentic orchestration offers a practical direction in this regard, allowing investigative reasoning to be decomposed into specialized cognitive roles rather than relying on monolithic prompting. Such modularity is especially important because cybercrime, evidence sources, and AI capabilities will continue to evolve, requiring architectures that can adapt to emerging technologies, new attack patterns, and future intelligent agents.

A particularly important implication of the findings is that knowledge grounding should be treated as a foundational requirement for forensic AI rather than an optional enhancement. The observed reduction in intent inflation across all evaluated models demonstrates that generic reasoning, even when technically competent, is insufficient for justice-sensitive investigative contexts. DF reasoning must operate within explicit legal and evidentiary boundaries, where technical suspiciousness alone does not establish criminal culpability. This aligns with broader concerns regarding trustworthy AI, explainability, and accountability in high-stakes decision environments [73], [75], [83], [84]. Structured knowledge sources, such as criminal knowledge bases, forensic threat intelligence repositories, and legal reference frameworks, can provide important process constraints that support evidentiary proportionality, hypothesis discipline, and attribution caution. Such knowledge

grounding also enables jurisdictional localization, allowing investigative reasoning to align with country-specific legal frameworks, procedural expectations, and technical threat landscapes. This becomes increasingly important as cyber-crime prosecution matures and DF outputs are more directly expected to support prosecutorial decision-making.

At the same time, the results reinforce that AI should function as an investigative assistant rather than an autonomous decision-maker. Although the proposed framework demonstrated meaningful analytical capability, forensic accountability remains fundamentally a human responsibility [81], [82]. DF conclusions can directly affect suspects, legal rights, investigative direction, and judicial outcomes, making unsupervised automation inappropriate. The appropriate role of AI is therefore to augment investigators through scalable reasoning support, evidentiary organization, alternative hypothesis generation, and decision assistance, while preserving human oversight for accountability, contextual judgment, and final legal interpretation.

Finally, the findings suggest that the future of DFs may require more than incremental digital transformation. The emergence of advanced LLMs, multimodal AI, autonomous cyber operations, and increasingly physical AI systems indicates that both cybercrime and forensic investigation are entering a period of discontinuous change rather than gradual evolution. Future DF systems will likely need to operate as adaptive intelligent ecosystems capable of integrating evolving AI models, specialized reasoning agents, legal knowledge sources, and emerging evidence domains. In this sense, the present work should be viewed not only as an investigation into IR, but as an early exploration of how legally aware agentic AI may contribute to the next generation of intelligent DFI systems.

This study has several limitations. First, the evaluation was conducted through structured scenario-based simulations rather than deployment within operational DF environments. While the experimental design enabled controlled comparative assessment across multiple crime types, intent categories, LLMs, and repeated executions, real-world investigations may introduce additional complexities such as incomplete evidence, ambiguous attribution, adversarial data manipulation, jurisdictional procedural constraints, and chain-of-custody considerations that were not explicitly modeled. Furthermore, the framework has not yet been independently evaluated by forensic practitioners or legal-domain experts, limiting claims of operational and legal validity.

Second, although the experimental scenarios were intentionally designed to represent diverse cybercrime contexts, including phishing, denial-of-service, and ransomware, they do not exhaust the broader landscape of cyber-enabled offenses. Additional crime categories, more complex multi-actor cases, insider threats, hybrid physical-digital incidents, and jurisdiction-specific legal variations may introduce reasoning challenges not fully captured in the present evaluation.

Third, the current implementation focuses on the analytical reasoning and recommendation phases of the broader

proposed DF framework, where IR is most directly operationalized. Earlier forensic phases such as evidence identification, acquisition, and examination were assumed to provide structured evidentiary inputs. Although this abstraction enabled focused evaluation of IR, future end-to-end implementations should assess the interaction between upstream evidence processing and downstream intent reasoning.

Although the CKB significantly improves reasoning discipline, the framework remains dependent on the capabilities and limitations of the underlying LLMs. As observed in recent studies of AI-assisted legal analysis, LLMs may generate plausible but unsupported conclusions and can exhibit varying levels of reliability across reasoning tasks [85]. Consequently, the proposed framework is intended to function as an intelligent investigative assistant rather than an autonomous decision-making system, with human experts retaining responsibility for evidentiary interpretation and legal judgment.

Finally, the implementation relies on contemporary LLMs, whose outputs may exhibit variability across executions and model versions. While repeated multi-model experimentation and structured knowledge guidance were introduced to improve robustness and reduce unsupported reasoning, model evolution and deployment-specific differences may influence performance over time. The proposed formal framework, however, remains implementation-independent and is not restricted to the current agentic LLM realization.

VIII. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

This study introduced a legally grounded Intention Recognition for Digital Forensics (IRDF) framework that integrates technical evidence interpretation with intent-based legal reasoning. The proposed framework provides a formal, implementation-independent representation of crimes, intent categories, evidentiary elements, actions, and states, enabling intention recognition to be incorporated as a structured component of digital forensic investigation. By extending forensic analysis beyond reconstructing what occurred to systematically assessing why actions were performed, the framework contributes a more legally informed approach to evidentiary interpretation and investigative reasoning.

To demonstrate the practical applicability of the framework, this study implemented an agentic AI realization of the IRDF model and evaluated it across multiple cybercrime scenarios, intent categories, LLMs, and repeated executions. The results indicate that structured knowledge-guided intention reasoning can substantially improve digital forensic analytical performance. In particular, the integration of the Criminal Knowledge Base improved intent classification accuracy while reducing unsupported inference inflation, resulting in more disciplined, explainable, and legally aligned reasoning behavior across models. These findings demonstrate the potential of knowledge-guided agentic AI to function as an effective forensic assistant for managing increasing investigative complexity while preserving necessary human oversight in legally sensitive decision-making contexts.

Future work should expand the IRDF framework through broader crime coverage, richer legal localization, and further refinement of its formal representations. From an implementation perspective, future research should explore end-to-end digital forensic realizations that incorporate evidence identification, collection, and examination processes, as well as alternative computational approaches beyond the current agentic LLM implementation. Additional studies using large real-world or benchmark forensic datasets, together with evaluation involving forensic practitioners and legal-domain experts, will be important for assessing the framework's scalability, robustness, investigative usefulness, legal alignment, and operational applicability under realistic forensic conditions. Further investigation is also needed to examine governance, ethical, explainability, and evidentiary considerations associated with the deployment of AI-assisted forensic decision-support systems.

REFERENCES

- [1] A. Kuzior, I. Tiutiunyk, A. Zielińska, and R. Kelemen, "Cybersecurity and cybercrime: Current trends and threats," *J. Int. Stud.*, vol. 17, no. 2, pp. 220–239, Jun. 2024.
- [2] V. Sharma, T. Manocha, S. Garg, S. Sharma, A. Garg, and R. Sharma, "Growth of cyber-crimes in society 4.0," in *Proc. 3rd Int. Conf. Innov. Practices Technol. Manage. (ICIPTM)*, Feb. 2023, pp. 1–6, doi: [10.1109/ICIPTM57143.2023.10118185](https://doi.org/10.1109/ICIPTM57143.2023.10118185).
- [3] D. S. Wall, *Cybercrime: The Transformation of Crime in the Information Age*, 2nd ed. Cambridge, U.K.: Polity Press, 2024.
- [4] J. Lusthaus, "Reconsidering crime and technology: What is this thing we call cybercrime?" *Annu. Rev. Law Social Sci.*, vol. 20, no. 1, pp. 369–385, Oct. 2024, doi: [10.1146/annurev-lawsocsci-041822-044042](https://doi.org/10.1146/annurev-lawsocsci-041822-044042).
- [5] S. R. Biedron, "Cybercrime in the digital age," Master's thesis, Centre Criminology, Fac. Law, Univ. Oxford, Oxford, U.K., 2024, doi: [10.5287/ora-r8oxybrng](https://doi.org/10.5287/ora-r8oxybrng).
- [6] H. N. Fakhouri, M. A. AlSharaiah, A. K. Alhwaitat, M. Alkaleleh, and F. F. Dweikat, "Overview of challenges faced by digital forensic," in *Proc. 2nd Int. Conf. Cyber Resilience (ICCR)*, Feb. 2024, pp. 1–8, doi: [10.1109/ICCR61006.2024.10532850](https://doi.org/10.1109/ICCR61006.2024.10532850).
- [7] A. MohanRaj Alenezi, "Digital forensics in the age of smart environments: A survey of recent advancements and challenges," 2023, *arXiv:2305.09682*.
- [8] D. Dunsin, M. C. Ghanem, K. Ouazzane, and V. Vassilev, "A comprehensive analysis of the role of artificial intelligence and machine learning in modern digital forensics and incident response," *Forensic Sci. Int., Digit. Invest.*, vol. 48, Mar. 2024, Art. no. 301675, doi: [10.1016/j.fsdi.2023.301675](https://doi.org/10.1016/j.fsdi.2023.301675).
- [9] G. Michelet, F. Breitingner, and G. Horsman, "Automation for digital forensics: Towards a definition for the community," *Forensic Sci. Int.*, vol. 349, Aug. 2023, Art. no. 111769, doi: [10.1016/j.forsciint.2023.111769](https://doi.org/10.1016/j.forsciint.2023.111769).
- [10] D. Quick and K.-K.-R. Choo, "Impacts of increasing volume of digital forensic data: A survey and future research challenges," *Digit. Invest.*, vol. 11, no. 4, pp. 273–294, Dec. 2014, doi: [10.1016/j.diin.2014.09.002](https://doi.org/10.1016/j.diin.2014.09.002).
- [11] A. A. Ahmed, N. Ahlami, and K. Zaman, "Attack intention recognition: A review," *Int. J. Netw. Secur.*, vol. 19, no. 2, pp. 244–250, 2017, doi: [10.6633/IJNS.201703.19\(2\).09](https://doi.org/10.6633/IJNS.201703.19(2).09).
- [12] T. Li, Y. Liu, Y. Liu, Y. Xiao, and N. A. Nguyen, "Attack plan recognition using hidden Markov and probabilistic inference," *Comput. Secur.*, vol. 97, Oct. 2020, Art. no. 101974, doi: [10.1016/j.cose.2020.101974](https://doi.org/10.1016/j.cose.2020.101974).
- [13] U. V. Naval Gund and K. Priyadarshini, "Crime intention detection system using deep learning," in *Proc. Int. Conf. Circuits Syst. Digit. Enterprise Technol. (ICCSDET)*, Dec. 2018, pp. 1–6, doi: [10.1109/ICCSDET.2018.8821168](https://doi.org/10.1109/ICCSDET.2018.8821168).
- [14] A. Årnes, Ed., *Digital Forensics*. Hoboken, NJ, USA: Wiley, 2018.
- [15] K. Kent, S. Chevalier, T. Grance, and H. Dang, "Guide to integrating forensic techniques into incident response," Nat. Inst. Standards Technol., Gaithersburg, MD, USA, Tech. Rep. NIST Special Publication 800-86, 2006.

- [16] N. A. Hassan, "Introduction: Understanding digital forensics," in *Digital Forensics Basics: A Practical Guide Using Windows OS*. Berkeley, CA, USA: Apress, 2019, pp. 1–33, doi: [10.1007/978-1-4842-3838-7_1](https://doi.org/10.1007/978-1-4842-3838-7_1).
- [17] T. J. Holt, A. M. Bossler, and K. C. Seigfried-Spellar, *Cybercrime and Digital Forensics: An Introduction*. London, U.K.: Routledge, 2015, doi: [10.4324/9781315777870](https://doi.org/10.4324/9781315777870).
- [18] A. K. Pandey, A. K. Tripathi, G. Kapil, V. Singh, M. W. Khan, A. Agrawal, R. Kumar, and R. A. Khan, "Current challenges of digital forensics in cyber security," in *Critical Concepts, Standards, and Techniques in Cyber Forensics*, M. S. Husain and M. Z. Khan, Eds., Hershey, PA, USA: IGI Global, 2020, pp. 31–46, doi: [10.4018/978-1-7998-1558-7.ch003](https://doi.org/10.4018/978-1-7998-1558-7.ch003).
- [19] R. Montasari, R. Hill, S. Parkinson, P. Peltola, A. Hosseinian-Far, and A. Daneshkhan, "Digital forensics: Challenges and opportunities for future studies," *Int. J. Organizational Collective Intell.*, vol. 10, no. 2, pp. 37–53, 2020, doi: [10.4018/ijoci.2020040103](https://doi.org/10.4018/ijoci.2020040103).
- [20] N. M. Karie and H. S. Venter, "Taxonomy of challenges for digital forensics," *J. Forensic Sci.*, vol. 60, no. 4, pp. 885–893, Jul. 2015, doi: [10.1111/1556-4029.12809](https://doi.org/10.1111/1556-4029.12809).
- [21] Y. W. Kassa, J. I. James, and E. G. Belay, "Cybercrime intention recognition: A systematic literature review," *Information*, vol. 15, no. 5, p. 263, May 2024, doi: [10.3390/info15050263](https://doi.org/10.3390/info15050263).
- [22] X. Zhang, H. Zhang, C. Li, P. Sun, Z. Liu, and J. Wang, "Network attack intention recognition based on signaling game model and netlogo simulation," in *Proc. Int. Conf. Digit. Soc. Intell. Syst. (DSIns)*, Dec. 2021, pp. 162–166, doi: [10.1109/dsins54396.2021.9670583](https://doi.org/10.1109/dsins54396.2021.9670583).
- [23] B. G. Bokolo, P. Onyehane, E. Ogegbene-Ise, I. Olufemi, and J. N. A. Tetey, "Leveraging machine learning for crime intent detection in social media posts," in *Proc. Int. Conf. AI-Generated Content*, 2023, pp. 224–236.
- [24] S. Abarna, J. I. Sheeba, S. Jayasrilakshmi, and S. P. Devaneyan, "Identification of cyber harassment and intention of target users on social media platforms," *Eng. Appl. Artif. Intell.*, vol. 115, Oct. 2022, Art. no. 105283, doi: [10.1016/j.engappai.2022.105283](https://doi.org/10.1016/j.engappai.2022.105283).
- [25] A. M. Bhugul and V. S. Gulhane, "Novel deep neural network for suspicious activity detection and classification," in *Proc. IEEE Int. Students' Conf. Electr. Electron. Comput. Sci. (SCEECS)*, Feb. 2023, pp. 1–7, doi: [10.1109/SCEECS57921.2023.10063130](https://doi.org/10.1109/SCEECS57921.2023.10063130).
- [26] N. Tsinganos, P. Fouliras, and I. Mavridis, "Leveraging dialogue state tracking for zero-shot chat-based social engineering attack recognition," *Appl. Sci.*, vol. 13, no. 8, p. 5110, 2023.
- [27] J. P. Bowen, "Gerard O'regan: Concise guide to formal methods: Theory, fundamentals and industry applications," *Formal Aspects Comput.*, vol. 32, no. 1, pp. 147–148, Feb. 2020, doi: [10.1007/s00165-020-00506-3](https://doi.org/10.1007/s00165-020-00506-3).
- [28] B. Weyers, J. Bowen, A. Dix, and P. Palanque, Eds., *The Handbook of Formal Methods in Human-Computer Interaction*. Cham, Switzerland: Springer, 2017, doi: [10.1007/978-3-319-51838-1](https://doi.org/10.1007/978-3-319-51838-1).
- [29] A. A. Ahmed and M. F. Mohammed, "SAIRF: A similarity approach for attack intention recognition using fuzzy min-max neural network," *J. Comput. Sci.*, vol. 25, pp. 467–473, Mar. 2018, doi: [10.1016/j.jocs.2017.09.007](https://doi.org/10.1016/j.jocs.2017.09.007).
- [30] D. Kim, D. Shin, D. Shin, and Y.-H. Kim, "Attack detection application with attack tree for mobile system using log analysis," *Mobile Netw. Appl.*, vol. 24, no. 1, pp. 184–192, Feb. 2019, doi: [10.1007/s11036-018-1012-4](https://doi.org/10.1007/s11036-018-1012-4).
- [31] X. Cheng, J. Zhang, and B. Chen, "Cyber situation comprehension for IoT systems based on APT alerts and logs correlation," *Sensors*, vol. 19, no. 18, p. 4045, Sep. 2019, doi: [10.3390/s19184045](https://doi.org/10.3390/s19184045).
- [32] R. Pang, X. Zhang, S. Ji, X. Luo, and T. Wang, "AdvMind: Inferring adversary intent of black-box attacks," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2020, pp. 1899–1907, doi: [10.1145/3394486.3403241](https://doi.org/10.1145/3394486.3403241).
- [33] R. Mirsky, Y. Shalom, A. Majadly, K. Gal, R. Puzis, and A. Felner, "New goal recognition algorithms using attack graphs," in *Proc. Int. Symp. Cyber Secur. Cryptogr. Mach. Learn.*, 2019, pp. 260–278.
- [34] B. Chen, Y. Liu, S. Li, and X. Gao, "Attack intent analysis method based on attack path graph," in *Proc. 9th Int. Conf. Commun. Netw. Secur.*, Nov. 2019, pp. 97–102, doi: [10.1145/3371676.3371680](https://doi.org/10.1145/3371676.3371680).
- [35] J. Zhao, X. Liu, Q. Yan, B. Li, M. Shao, H. Peng, and L. Sun, "Automatically predicting cyber attack preference with attributed heterogeneous attention networks and transductive learning," *Comput. Secur.*, vol. 102, Mar. 2021, Art. no. 102152, doi: [10.1016/j.cose.2020.102152](https://doi.org/10.1016/j.cose.2020.102152).
- [36] J. Kang, H. Yang, Y. Zhang, Y. Dai, M. Zhan, and W. Wang, "ActDetector: A sequence-based framework for network attack activity detection," in *Proc. IEEE Symp. Comput. Commun. (ISCC)*, Jun. 2022, pp. 1–7, doi: [10.1109/ISCC55528.2022.9912824](https://doi.org/10.1109/ISCC55528.2022.9912824).
- [37] T.-Y. Hsu and C.-Y. Tang, "Detection of malicious activities using machine learning in physical environments," in *Proc. Int. Conf. Comput. Sci. Comput. Intell. (CSCI)*, Dec. 2022, pp. 1047–1052, doi: [10.1109/CSCI58124.2022.00185](https://doi.org/10.1109/CSCI58124.2022.00185).
- [38] K. Guang, T. Guangming, D. Xia, W. Shuo, and W. Kun, "A network security situation assessment method based on attack intention perception," in *Proc. 2nd IEEE Int. Conf. Comput. Commun. (ICCC)*, Oct. 2016, pp. 1138–1142, doi: [10.1109/CompComm.2016.7924882](https://doi.org/10.1109/CompComm.2016.7924882).
- [39] G. A. Martínez-Mascorro, J. R. Abreu-Pederzini, J. C. Ortiz-Bayliss, A. García-Collantes, and H. Terashima-Marín, "Criminal intention detection at early stages of shoplifting cases by using 3D convolutional neural networks," *Computation*, vol. 9, no. 2, p. 24, Feb. 2021, doi: [10.3390/computation9020024](https://doi.org/10.3390/computation9020024).
- [40] R. Pandey, H. Purohit, B. Stabile, and A. Grant, "Distributional semantics approach to detect intent in Twitter conversations on sexual assaults," in *Proc. IEEE/WIC/ACM Int. Conf. Web Intell. (WI)*, Dec. 2018, pp. 270–277, doi: [10.1109/WI.2018.00-80](https://doi.org/10.1109/WI.2018.00-80).
- [41] N. Tsinganos, P. Fouliras, and I. Mavridis, "Applying BERT for early-stage recognition of persistence in chat-based social engineering attacks," *Appl. Sci.*, vol. 12, no. 23, p. 12353, Dec. 2022, doi: [10.3390/app122312353](https://doi.org/10.3390/app122312353).
- [42] A. Wickramasekara, F. Breiteringer, and M. Scanlon, "Exploring the potential of large language models for improving digital forensic investigation efficiency," *Forensic Sci. Int., Digit. Invest.*, vol. 52, Mar. 2025, Art. no. 301859, doi: [10.1016/j.fsidi.2024.301859](https://doi.org/10.1016/j.fsidi.2024.301859).
- [43] E. Xu, W. Zhang, and W. Xu, "Transforming digital forensics with large language models: Unlocking automation, insights, and justice," in *Proc. 33rd ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2024, pp. 5543–5546, doi: [10.1145/3627673.3679091](https://doi.org/10.1145/3627673.3679091).
- [44] H. Henseler and H. van Beek, "ChatGPT as a copilot for investigating digital evidence," in *Proc. LegalAIA@ ICAIL*, 2023, pp. 58–69.
- [45] M. N. Rahman, T. Mohammad, and S. Virtanen, "Leveraging LLMs for network traffic analysis: Design, implementation, and evaluation of an LLM-powered system for cyber incident reconstruction," M.S. thesis, Univ. Turku, Turku, Finland, 2024.
- [46] H. Zhou, W. Xu, J. Dehlinger, S. Chakraborty, and L. Deng, "An LLM-driven approach to gain cybercrime insights with evidence networks," in *Proc. Usenix Workshop*, 2024, pp. 1–3.
- [47] O. G. Lira, A. Marroquin, and M. A. To, "Harnessing the advanced capabilities of llm for adaptive intrusion detection systems," in *Proc. Int. Conf. Adv. Inf. Netw. Appl.*, 2024, pp. 453–464.
- [48] K.-J. Kim, C.-H. Lee, S.-E. Bae, J.-H. Choi, and W. Kang, "Digital forensics in law enforcement: A case study of LLM-driven evidence analysis," *Forensic Sci. Int., Digital Investigation*, vol. 54, Sep. 2025, Art. no. 301939, doi: [10.1016/j.fsidi.2025.301939](https://doi.org/10.1016/j.fsidi.2025.301939).
- [49] G. Michelet and F. Breiteringer, "ChatGPT, llama, can you write my report? An experiment on assisted digital forensics reports written using (local) large language models," *Forensic Sci. Int., Digit. Invest.*, vol. 48, Mar. 2024, Art. no. 301683, doi: [10.1016/j.fsidi.2023.301683](https://doi.org/10.1016/j.fsidi.2023.301683).
- [50] A. Gill, "What is crime mapping," in *Government Office for London*, vol. 1. London, U.K., 2013, pp. 1–4.
- [51] S. Bell, "Crime," in *A Dictionary of Forensic Science*. Oxford, U.K.: Oxford University Press, 2013, doi: [10.1093/acref/9780199594009.013.0265](https://doi.org/10.1093/acref/9780199594009.013.0265).
- [52] G. Gooch and M. Williams, "Crime," in *A Dictionary of Law Enforcement*. Oxford, U.K.: Oxford University Press, 2015, doi: [10.1093/acref/9780191758256.013.0805](https://doi.org/10.1093/acref/9780191758256.013.0805).
- [53] Oxford Dictionaries, "Crime," in *The Oxford Essential Dictionary of the U.S. Military*. New York, NY, USA: Oxford University Press, 2002, doi: [10.1093/acref/9780199891580.013.2071](https://doi.org/10.1093/acref/9780199891580.013.2071).
- [54] Federal Democratic Republic of Ethiopia, *Computer Crime Proclamation No.958/2016*, Addis Ababa, Ethiopia, 2016, p. 9104.
- [55] American Law Institute, *Model Penal Code: Official Draft and Explanatory Notes: Complete Text of Model Penal Code as Adopted at the 1962 Annual Meeting of the American Law Institute at Washington, D.C., May 24, 1962*. Philadelphia, PA, USA: American Law Institute, 1985.
- [56] G. Gooch and M. Williams, "Intention," in *A Dictionary of Law Enforcement*. Oxford, U.K.: Oxford University Press, 2015, doi: [10.1093/acref/9780191758256.013.1671](https://doi.org/10.1093/acref/9780191758256.013.1671).

- [57] J. Law, "Intention," in *A Dictionary of Law*, 10th ed. Oxford, U.K.: Oxford University Press, 2022, doi: [10.1093/acref/9780192897497.013.2003](https://doi.org/10.1093/acref/9780192897497.013.2003).
- [58] M. Shahin, "Criminal intention and motive in criminal law: A comparative approach," in *Proc. Researchgate*, pp. 1–5, Jul. 2021, doi: [10.13140/RG.2.2.21341.33766](https://doi.org/10.13140/RG.2.2.21341.33766).
- [59] G. Coffey, "Codifying the meaning of 'intention' in the criminal law," *J. Criminal Law*, vol. 73, no. 5, pp. 394–413, Oct. 2009, doi: [10.1350/jcla.2009.73.5.590](https://doi.org/10.1350/jcla.2009.73.5.590).
- [60] T. Masterman, S. Besen, M. Sawtell, and A. Chao, "The landscape of emerging AI agent architectures for reasoning, planning, and tool calling: A survey," 2024, *arXiv:2404.11584*.
- [61] N. Krishnan, "AI agents: Evolution, architecture, and real-world applications," 2025, *arXiv:2503.12687*.
- [62] C. Hargreaves, H. van Beek, and E. Casey, "SOLVE-IT: A proposed digital forensic knowledge base inspired by MITRE ATT&CK," *Forensic Sci. Int., Digit. Invest.*, vol. 52, Mar. 2025, Art. no. 301864, doi: [10.1016/j.fsidi.2025.301864](https://doi.org/10.1016/j.fsidi.2025.301864).
- [63] G. Palmer, "A road map for digital forensic research," in *Proc. First Digital Forensic Research Workshop (DFRWS)*, Utica, NY, USA, 2001, pp. 27–30.
- [64] B. E. Strom, A. Applebaum, D. P. Miller, K. C. Nickels, A. G. Pennington, and C. B. Thomas, "MITRE ATT&CK: Design and philosophy," The MITRE Corporation, McLean, VA, USA, Tech. Rep. MP180360, 2018.
- [65] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.
- [66] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, "A design science research methodology for information systems research," *J. Manag. Inf. Syst.*, vol. 24, no. 3, pp. 45–77, 2007, doi: [10.2753/MIS0742-1222240302](https://doi.org/10.2753/MIS0742-1222240302).
- [67] T. B. Brown et al., "Language models are few-shot learners," in *Proc. NeurIPS*, 2020, pp. 1–25.
- [68] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 24824–24837.
- [69] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," in *Proc. ICLR*, 2023, pp. 1–24.
- [70] Z. Xi et al., "The rise and potential of large language model based agents: A survey," 2023, *arXiv:2309.07864*.
- [71] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu, A. Hassan Awadallah, R. W. White, D. Burger, and C. Wang, "AutoGen: Enabling next-gen LLM applications via multi-agent conversation," 2023, *arXiv:2308.08155*.
- [72] J. S. Park, J. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proc. 36th Annu. ACM Symp. User Interface Softw. Technol.*, Oct. 2023, pp. 1–22.
- [73] F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," 2017, *arXiv:1702.08608*.
- [74] W. Xin Zhao et al., "A survey of large language models," 2023, *arXiv:2303.18223*.
- [75] A. B. Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, S. Garcia, S. Gil-Lopez, D. Molina, R. Benjamins, R. Chatila, and F. Herrera, "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Inf. Fusion*, vol. 58, pp. 82–115, Jun. 2020.
- [76] E. Casey, *Digital Evidence and Computer Crime: Forensic Science, Computers and the Internet*. New York, NY, USA: Academic, 2011.
- [77] G. Horsman, "Raiders of the lost artefacts: Championing the need for digital forensics research," *Forensic Sci. Int., Rep.*, vol. 1, Nov. 2019, Art. no. 100003.
- [78] R. Resende de Mendonça, D. Felix de Brito, F. de Franco Rosa, J. C. dos Reis, and R. Bonacin, "A framework for detecting intentions of criminal acts in social media: A case study on Twitter," *Information*, vol. 11, no. 3, p. 154, Mar. 2020.
- [79] E. Casey, S. Barnum, R. Griffith, J. Snyder, H. van Beek, and A. Nelson, "Advancing coordinated cyber-investigations and tool interoperability using a community developed specification language," *Digit. Invest.*, vol. 22, pp. 14–45, Sep. 2017.
- [80] M. M. Pollitt, "A history of digital forensics," in *Advances in Digital Forensics VI*, vol. 337, K.-P. Chow and S. Sheno, Eds., Berlin, Germany: Springer, 2010, pp. 3–15, doi: [10.1007/978-3-642-15506-2_1](https://doi.org/10.1007/978-3-642-15506-2_1).
- [81] *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, National Institute of Standards and Technology, Gaithersburg, MD, USA, 2023.
- [82] European Parliament and Council of the European Union, "Regulation (EU) 2024/1689 of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)," *Off. J. Eur. Union, L 2024/1689*, Jul. 2024.
- [83] K. D. Ashley, *Artificial Intelligence and Legal Analytics: New Tools for Law Practice in the Digital Age*. Cambridge, U.K.: Cambridge Univ. Press, 2017.
- [84] T. Bench-Capon, "Using issues to explain legal decisions," *Artif. Intell. Law*, vol. 29, no. 4, pp. 563–588, 2021.
- [85] L. Peoples, "Artificial intelligence and legal analysis: Implications for legal education and the profession," 2025, *arXiv:2502.03487*.

YIDNEKACHEW WORKU KASSA received the B.Sc. degree in software engineering from the MicroLink Information Technology College, Addis Ababa, Ethiopia, in 2004, and the M.Sc. degree in computer science from the HiLCoE School of Computer Science and Technology, Addis Ababa, in 2019. He is currently pursuing the Ph.D. degree in cybersecurity and artificial intelligence with Addis Ababa University, Addis Ababa.

He has extensive professional experience in the field of information security, where he progressed from a Junior Developer to senior technical and leadership roles with the Information Network Security Agency (INSA), Ethiopia. From 2018 to 2022, he was the Deputy Director General of the agency. He is also a State Minister. He has co-authored a publication titled *Cybercrime IR: A Systematic Literature Review*. His research interests include artificial intelligence and cybersecurity.

ELEFELIOUS GETACHEW BELAY received the B.Sc. degree in computer science from Addis Ababa University, the M.Sc. degree in software engineering from the Vrije University of Amsterdam, The Netherlands, and the Ph.D. degree in software engineering from Addis Ababa University (AAU).

He is an Associate Professor of software engineering with over 15 years of experience in industry, academia, and research. He was also a Visiting Researcher with Virginia Tech, USA; and a Post-Doctoral Researcher with the University of Milan, Italy. He has an extensive experience in the industry working in various capacities from a Senior Expert to the Director level positions. Currently, he is the Lead Researcher with the Center of Artificial Intelligence and Robotics, Institute of Advanced Science and Technology, AAU.

Dr. Belay has received several research awards from Facebook, IDRC/Carleton University, Research ICT Africa, and AAU. He also received the JP Morgan Faculty Research Award, in 2021/2022.

JOSHUA I. JAMES received the Ph.D. degree in digital forensic investigation from University College Dublin. His doctoral research explored automated human inference in digital forensic investigations using hypothesis reduction, contributing foundational work in formalizing digital investigative reasoning.

He is a Digital Forensics Researcher, an Educator, and a Practitioner whose work focuses on digital forensic science (DFIR), cybercrime investigations, automation of forensic reasoning, and international cooperation in digital evidence. He is the Founder of DFIR Science LLC, a platform dedicated to advancing digital forensic research, education, and professional practice. He was an Associate Professor with the Digital Forensic Investigation Research (DFIRE) Laboratory, Hallym University, and has taught and conducted research in digital investigations, cybercrime, and forensic automation.

...